

finds that the stocks with the high tax burdens have average returns 2.5% higher, after controlling for the Fama–French effects, than stocks with low tax burdens.

The fact that taxes affect expected returns does not mean that the clientele sorting predicted by Miller and Scholes (1978)—where different types of investors are drawn to stocks with high or low dividend yields—does not occur. Institutions prefer to own stocks that pay dividends and individual investors are drawn to non-dividend paying stocks. Graham and Kumar (2006) show, however, that while taxes do induce clientele effects in individual investors, the effects are much smaller than non-tax-based investor characteristics such as age and wealth.

The average, representative investor is affected by taxes, so tax as a factor is ideally exploited by a tax-exempt investor. By definition, individuals buying stocks with high tax yields will be paying more tax. But tax-exempt investors will not, and they should use their comparative advantage.

6. Pre-Tax and After-Tax Returns Redux

Duncan's accountant is right in emphasizing after-tax returns over pre-tax returns. The latter are relevant for holding assets in a taxable account. But Duncan may be able to hold the assets in a tax-deferred account. Thus Duncan faces both an asset allocation problem and an asset location problem.

The overriding rule is to maximize savings in tax-deferred vehicles first and to hold assets with the most onerous tax treatments (corporate bonds and high dividend paying stocks) in tax-deferred vehicles. This aligns with popular industry advice, but to do it properly requires computing an adjustment to convert tax-deferred wealth to taxable wealth so that a standard asset allocation problem can be solved over taxable wealth. The optimal asset location decision implies we should not observe the same mix of stocks and bonds in tax-deferred and taxable accounts. Sadly, most investors are very far from being tax efficient.

Taxes are a factor for asset prices. The fact that muni bonds have lower yields, on average, is evidence that asset prices reflect tax effects. Munis have historically had little credit risk. Muni yield curves have never sloped downward and are more steeply sloped than Treasuries. But munis are extremely illiquid and the market is hugely inefficient.

Taxes also affect equity prices. Stocks with high tax burdens have higher returns than stocks with low tax burdens, in excess of those predicted by the standard value/growth, momentum, and other factor premiums.